

Practical 7

Code:

```
polisettyreesharsha@SREEHA x + v
GNU nano 8.7.1                                     Prac7code.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_var = 10;
static int static_var = 20;

void sample_function()
{
    printf("Code address   : %p\n", (void *)sample_function);
}

int main()
{
    int stack_var = 30;
    int *heap_var = malloc(sizeof(int));

    if (heap_var == NULL)
    {
        printf("Memory allocation failed.\n");
        return 1;
    }

    *heap_var = 40;

    printf("Process ID      : %d\n\n", getpid());

    sample_function();

    printf("Global address : %p\n", (void *)&global_var);
    printf("Static address : %p\n", (void *)&static_var);
    printf("Heap address  : %p\n", (void *)heap_var);
    printf("Stack address  : %p\n", (void *)&stack_var);

    printf("\nMemory mappings from /proc/%d/maps:\n\n", getpid());

    char command[100];

    sample_function();

    printf("Global address : %p\n", (void *)&global_var);
    printf("Static address : %p\n", (void *)&static_var);
    printf("Heap address  : %p\n", (void *)heap_var);
    printf("Stack address  : %p\n", (void *)&stack_var);

    printf("\nMemory mappings from /proc/%d/maps:\n\n", getpid());

    char command[100];
    snprintf(command, sizeof(command), "cat /proc/%d/maps", getpid());

    system(command);

    free(heap_var);

    return 0;
}
```

Output:

```
polisettyreearsha@SREEHA x + v
polisettyreearsha@SREEHARSHA:~$ nano Prac7code.c
polisettyreearsha@SREEHARSHA:~$ gcc Prac7code.c -o Prac7code
polisettyreearsha@SREEHARSHA:~$ ./Prac7code
Process ID      : 539

Code address    : 0x64aa94512229
Global address  : 0x64aa94515010
Static address  : 0x64aa94515014
Heap address    : 0x64aa95724010
Stack address   : 0x7ffe6726f5e4

Memory mappings from /proc/539/maps:

64aa94511000-64aa94512000 r--p 00000000 08:30 22337 /home/polisettyreearsha/Prac7code
64aa94512000-64aa94513000 r-xp 00001000 08:30 22337 /home/polisettyreearsha/Prac7code
64aa94513000-64aa94514000 r--p 00002000 08:30 22337 /home/polisettyreearsha/Prac7code
64aa94514000-64aa94515000 r--p 00002000 08:30 22337 /home/polisettyreearsha/Prac7code
64aa94515000-64aa94516000 rw-p 00003000 08:30 22337 /home/polisettyreearsha/Prac7code
64aa95724000-64aa95724500 rw-p 00000000 00:00 0 [heap]
704ecee00000-704ecee28000 r--p 00000000 08:30 13647 /usr/lib/x86_64-linux-gnu/libc.so.6
704ecee28000-704ecfc0000 r-xp 00028000 08:30 13647 /usr/lib/x86_64-linux-gnu/libc.so.6
704ecfc00000-704ecf00e000 r--p 001c0000 08:30 13647 /usr/lib/x86_64-linux-gnu/libc.so.6
704ecf00e000-704ecf012000 r--p 0020d000 08:30 13647 /usr/lib/x86_64-linux-gnu/libc.so.6
704ecf012000-704ecf014000 rw-p 00211000 08:30 13647 /usr/lib/x86_64-linux-gnu/libc.so.6
704ecf014000-704ecf021000 rw-p 00000000 00:00 0
704ecf021000-704ecf02b000 rw-p 00000000 00:00 0
704ecf02b000-704ecf0a2000 rw-p 00000000 00:00 0
704ecf0a2000-704ecf0a6000 r--p 00000000 00:00 0 [vvar]
704ecf0a6000-704ecf0a8000 r--p 00000000 00:00 0 [vvar_vclock]
704ecf0a8000-704ecf0aa000 r-xp 00000000 00:00 0 [vdso]
704ecf0aa000-704ecf0ab000 r--p 00000000 08:30 13644 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
704ecf0ab000-704ecf0da000 r-xp 00001000 08:30 13644 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
704ecf0da000-704ecf0e5000 r--p 00030000 08:30 13644 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
704ecf0e5000-704ecf0e7000 r--p 0003b000 08:30 13644 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
704ecf0e7000-704ecf0e8000 rw-p 0003d000 08:30 13644 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
704ecf0e8000-704ecf0e9000 rw-p 00000000 00:00 0
7ffe67250000-7ffe67271000 rw-p 00000000 00:00 0 [stack]
polisettyreearsha@SREEHARSHA:~$
```